

A Canonical Biorthogonal Spectral Packet

Balanced Coordinates on Source–Observation Riesz Channels

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Abstract

RH-195 associates each selected simple physical eigenvalue with exact right and left matrix states $X_i = P_i S$ and $Y_i = P_i^* O^*$ whose cross pairing is diagonal with entries equal to transfer residues. This paper constructs optimal packet coordinates on the resulting right and left spectral spaces.

Let $E_R = \text{span}\{X_i\}$ and $E_L = \text{span}\{Y_i\}$, assume every residue is nonzero, and let Q_R, Q_L be orthonormal bases. If $Q_L^* Q_R = U \Sigma V^*$, define

$$V_c = Q_R V \Sigma^{-1/2}, \quad W_c = Q_L U \Sigma^{-1/2}.$$

Then $W_c^* V_c = I$, both frame norms equal $\sigma_{\min}^{-1/2}$, and their norm product $\sigma_{\min}^{-1}$ is optimal among all biorthogonal coordinates on the same two spaces.

Because E_R and E_L are exact invariant spaces,

$$\mathcal{L}_A V_c = V_c K_c, \quad \mathcal{L}_A^* W_c = W_c K_c^*.$$

Both directed residuals vanish, K_c is similar to the selected physical spectral restriction, and its determinant and every power trace are exact. A 140-case complex nonnormal audit verifies all identities with zero failures.

The packet is canonical relative to the chosen Riesz contours, source, and observation. It removes the complement self-energy inside that selected channel. Intrinsic contour selection, interval validation, and uniform cross-level transversality remain open.

1 From residue coordinates to balanced coordinates

For pairwise disjoint simple eigenvalues, RH-195 gives right and left channel states

$$X_i = P_i S, \quad Y_i = P_i^* O^*, \tag{1}$$

with

$$\langle Y_i, X_j \rangle_F = \delta_{ij} r_i. \tag{2}$$

If $r_i \neq 0$, the direct normalization $V_i = X_i$, $W_i = Y_i / \bar{r}_i$ is biorthogonal. It preserves mode labels and diagonalizes the compressed operator, but it need not balance coordinate norms [2]. The cross-Gram balancing used below is the exact spectral specialization of the temporal construction in RH-184 [1].

The geometry depends only on the two subspaces

$$E_R = \text{span}\{X_1, \dots, X_r\}, \quad E_L = \text{span}\{Y_1, \dots, Y_r\}. \quad (3)$$

Nonzero residues imply that the pairing between them is nondegenerate.

2 Cross-Gram geometry

Choose Frobenius-orthonormal bases $Q_R, Q_L \in \mathbb{C}^{N \times r}$ for the vectorized spaces and form

$$H = Q_L^* Q_R. \quad (4)$$

The singular values of H are the cosines of the principal angles between E_L and E_R . Transversality is equivalent to $\sigma_{\min}(H) > 0$.

Let

$$H = U \Sigma V^*. \quad (5)$$

Theorem 2.1 (Balanced spectral packet). *Define*

$$V_c = Q_R V \Sigma^{-1/2}, \quad W_c = Q_L U \Sigma^{-1/2}. \quad (6)$$

Then

$$W_c^* V_c = I, \quad (7)$$

and

$$\|V_c\| = \|W_c\| = \sigma_{\min}(H)^{-1/2}. \quad (8)$$

Proof. Insert the singular-value decomposition into $W_c^* V_c$. Since Q_R, Q_L, U, V are isometries, the operator norms of the frames equal the largest diagonal entry of $\Sigma^{-1/2}$. $\square$

3 Optimality

Any frames V, W spanning E_R, E_L and satisfying $W^* V = I$ define the same oblique projector $\Pi = V W^*$ from the ambient space onto E_R along $E_L^\perp$. Its norm is

$$\|\Pi\| = \sigma_{\min}(H)^{-1}. \quad (9)$$

Theorem 3.1 (Optimal norm product). *Every biorthogonal realization on (E_R, E_L) obeys*

$$\|V\| \|W\| \geq \sigma_{\min}(H)^{-1}. \quad (10)$$

The balanced frames attain equality. They also minimize $\max(\|V\|, \|W\|)$, whose optimal value is $\sigma_{\min}(H)^{-1/2}$.

Proof. Submultiplicativity gives $\|V W^*\| \leq \|V\| \|W\|$, while the left side equals $\sigma_{\min}^{-1}$. The balanced norms attain the lower bound. For any positive numbers with product at least c , their maximum is at least $\sqrt{c}$; scalar gauge balancing attains this value. $\square$

The potentially large conditioning is intrinsic to the source-observation subspaces. No alternative basis can remove it.

4 Exact two-sided invariance

The right space is invariant under $\mathcal{L}_A : X \mapsto AX$, and the left space is invariant under $\mathcal{L}_A^*$. Hence there are matrices K_R, K_L with

$$\mathcal{L}_A V_c = V_c K_R, \quad \mathcal{L}_A^* W_c = W_c K_L. \quad (11)$$

Theorem 4.1 (One exact compressed operator). *With*

$$K_c = W_c^* \mathcal{L}_A V_c, \quad (12)$$

one has

$$\boxed{\mathcal{L}_A V_c = V_c K_c, \quad \mathcal{L}_A^* W_c = W_c K_c^*}. \quad (13)$$

Thus both directed Ritz residuals vanish exactly.

Proof. The first invariant relation and $W_c^* V_c = I$ imply $K_R = K_c$. Taking the adjoint of the left invariant relation gives $W_c^* \mathcal{L}_A = K_L^* W_c^*$. Multiplying by V_c shows $K_L^* = K_c$, hence the second identity. $\square$

This is the exact endpoint that the temporal bi-Krylov packet approximates. In these coordinates the packet-to-complement couplings are zero because the selected subspaces are reducing in the biorthogonal sense.

5 Spectrum, determinant, and traces

In the residue-labeled frame, the compressed matrix is $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_r)$. Every other biorthogonal frame on the same spaces differs by a gauge $V \mapsto VG$, $W \mapsto W(G^{-1})^*$, so

$$K \mapsto G^{-1} K G. \quad (14)$$

Corollary 5.1 (Exact finite spectral ledger). *For the balanced packet,*

$$\det(zI - K_c) = \prod_{j=1}^r (z - \lambda_j), \quad (15)$$

$$\text{tr } K_c^q = \sum_{j=1}^r \lambda_j^q \quad (q \geq 1). \quad (16)$$

These are exact finite channel identities. They are not the determinant or trace of the complete Frobenius operator, whose multiplicities remain multiplied by m .

6 Canonicity and its limits

The spaces E_R, E_L are determined by (A, S, O) and the selected Riesz contours. The balanced packet is unique up to unitary transformations within repeated singular-value blocks and harmless phase conventions. Its determinant and traces are fully invariant under this residual gauge.

This is a useful but conditional canonicity. If the contours were selected by first inspecting the target eigenvalues, the construction has not yet explained why those modes are dynamically privileged. Gate A needs an a priori selection principle, for example an edge-gap, source-history, or renormalization rule that survives refinement.

7 Identity audit

The numerical audit uses 140 complex nonnormal systems with base dimensions 4–10 and source widths 1–4. Four spectral channels of largest modulus are formed from exact computed left/right eigendata. The code balances their right and left matrix-state spaces and verifies:

1. equality of balanced frame norms;
2. biorthogonality;
3. zero right and left invariant residuals;
4. equality of compressed and selected spectra;
5. Newton power-trace identities through power eight.

All cases pass the declared 10^{-8} tolerance.

8 Position in the route

The local route now contains an exact endpoint:

temporal packet $\longrightarrow$ matched Riesz channels $\longrightarrow$ balanced exact packet.

The first arrow is finite and numerical; the second is exact once the contours are fixed. RH-197 measures the physical residues and the intrinsic cross-angle condition of the matched quartet. RH-198 then quantifies how the temporal packet approaches this exact endpoint.

No statement here constructs a self-adjoint operator, a Hilbert–Pólya spectrum, a prime-power trace formula, or a zeta divisor. The work remains inside the physical spectral interface portion of Gate A.

References

- [1] Liang Wang. Balanced biorthogonal temporal realization, 2026. Prime Dynamics Theory Program, RH-184.
- [2] Liang Wang. Source–observation riesz channels, 2026. Prime Dynamics Theory Program, RH-195.